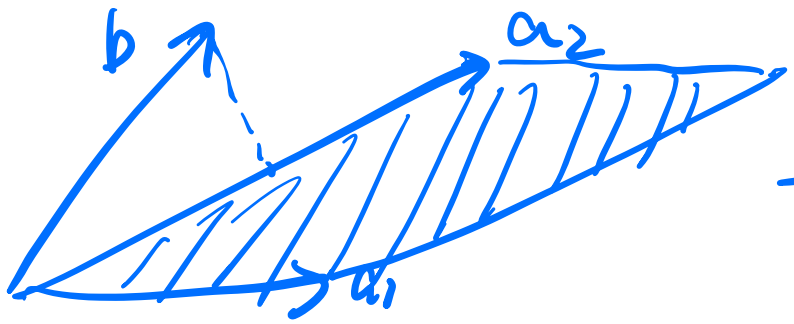
projection:

$p = \hat{x}_1 a_1 + \hat{x}_2 a_2$

$= A\hat{x}$



find a combination of a_1 a_2 s.t.

$\vec{b} - \vec{p} \perp \text{col}(A)$

$b - A\hat{x} \perp \text{col}(A)$

$a_1^T(b - A\hat{x}) = 0$

$a_2^T(b - A\hat{x}) = 0$

$$\begin{bmatrix} — a_1^T — \\ — a_2^T — \end{bmatrix} \begin{bmatrix} b - A\hat{x} \end{bmatrix}$$

$\rightarrow A^T(b - A\hat{x}) = 0$

exactly

$a^T(b - x\vec{a}) = 0$

e in $\text{Null}(A^T)$

↓

$e \perp \text{col}(A)$!

$A^T A\hat{x} = A^T b$

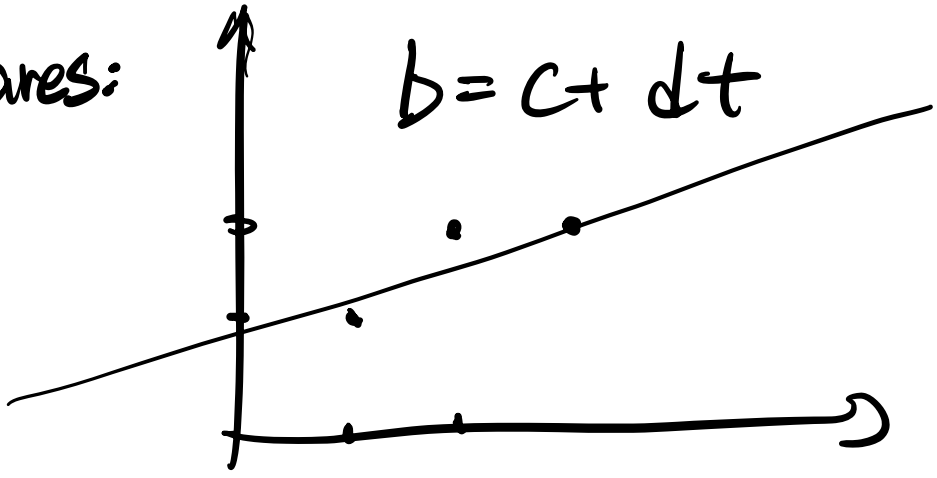
$\hat{x} = (A^T A)^{-1} A^T b$

projection: $A\hat{x} = \underbrace{A(A^T A)^{-1} A^T}_{\text{projection}} b$

$P = A(A^T A)^{-1} A^T b$

$$\begin{cases} \text{col}(P) = \text{col}(A) \\ P = P^T \\ P^2 = P \end{cases}$$

Least Squares: $b = C + dt$



cannot solve

$$Ax = b$$

solve $Ax = A(A^TA)^{-1}A^Tb$

$$A^TAx = A^TA(A^TA)^{-1}A^Tb$$

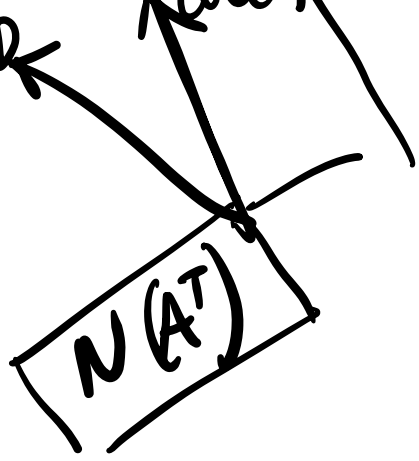
$$A^TAx = A^Tb$$

$$\begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} C \\ d \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$$

x $\qquad$ y

P(b)
col(A)

P is projection



$$b = p + e$$

$$e = b(I-P)$$

↓ projection onto $N(A^T)$

$$\|Ax-b\| = e^2$$

$$= e_1^2 + e_2^2 + e_3^2$$

$$= (c+d-1)^2 + (c+2d-2)^2 + (c+3d-2)^2$$

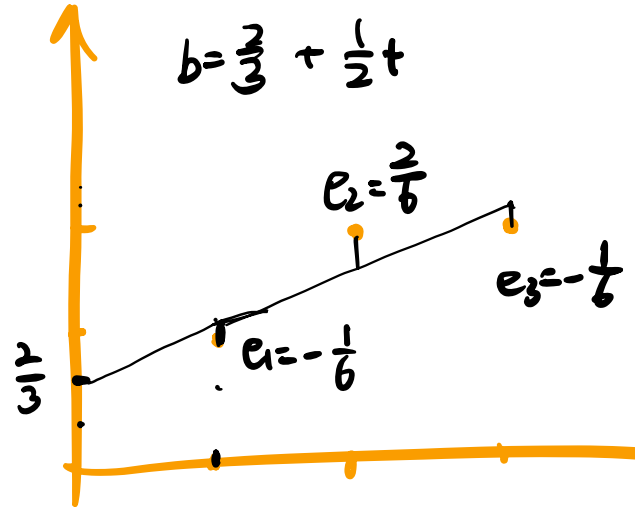
$$\frac{de}{dc} = 6c + 2(d-1) + 2(2d-2) + 2(3d-2)$$

→ $6c + 12d = 10$

$$\begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} c \\ d \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$$

$$\left[\begin{array}{cc|c} 3 & 6 & 5 \\ 6 & 14 & 11 \end{array}\right]$$

$$c = \frac{2}{3} \quad d = \frac{1}{2}$$



$$b = c + dt$$

$$= \frac{2}{3} + \frac{1}{2}t$$

$$p = \begin{bmatrix} \frac{7}{6} \\ \frac{10}{6} \\ \frac{13}{6} \end{bmatrix} \quad e = \begin{bmatrix} -\frac{1}{6} \\ \frac{0}{6} \\ -\frac{1}{6} \end{bmatrix}$$

$p \perp e$!

Lemma If A has independent col.

A^TA is invertible

$A^TAx=0 \xrightarrow{\text{wts}} x=0$

Pf: $x^TA^TAx=0$

$(Ax)^T(Ax)=0$

$\Rightarrow \quad Ax=0$

But A has independent columns!

$x=0$ $\square$.

Gram-Schmidt

Symmetric

$A^T = A$

$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

in between.

Anti-symmetric

$A^T = -A$

$\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$